

- 37** Some animals have genes that code for small peptides called cathelicidins. These peptides kill a wide range of bacteria by attaching to lipids in bacterial membranes, so weakening or disrupting them.

Scientists have produced a synthetic version of the cathelicidin that kills bacteria that are resistant to a number of antibiotics such as tetracycline.

Which pair of statements explain how this synthetic cathelicidin might help with the problem of antibiotic resistance?

- 1 It is synthetic so bacteria can never become resistant to it.
- 2 It could be used instead of tetracycline, allowing tetracycline resistance to be reduced.
- 3 The only way a bacterium could develop resistance to it is by altering all the lipids in its membranes.
- 4 It could be used to kill multidrug-resistant strains of bacteria for which there is no effective antibiotic.

A 1 and 3

B 1 and 4

C 2 and 3

D 2 and 4